Report — Proportional Control for Maneuvering Flapping Flight

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Contents

1 Summary

- As long as body velocity is normal to the stroke plane, our model reduces the wingbeat-averaged aerodynamic force to a product $F_a^* = q^* C_{F^*}$, with a magnitude scale q^* set by flapping amplitude and frequency, and a force coefficient C_{F^*} set by the wing pitch kinematics and advance ratio.
- Force direction relative to the stroke plane normal is controlled solely by the mean pitch ψ_0
- Two control handles follow for a given stroke plane angle: flapping amplitude ϕ_1 sets force magnitude and the mean pitch ψ_0 sets its direction
- Using the above variables and tuning a controller to run an order of magnitude below the wingbeat frequency, we hold a stable hover across the full morphological parameter range and for different stroke plane angles
- Inclined stroke planes are found to have greater damping in the vertical direction
- With the quasi-steady aerodynamic model, an inclined stroke plane carries no efficiency benefit for hover. Nonetheless we use an inclined stroke plane (hover stroke plane angle is a free parameter) in the control scheme to mimic dragonfly wing kinematics
- The hover controller is extended to all-direction flight by slaving the stroke plane angle γ and the wing pitch amplitude ψ_1 to the body velocity
- In unsteady closed-loop simulation the controller tracks accelerations, a pull-up, a fully reversing flight loop, and pure-pursuit prey capture. It is limited in sustained descents.

2 Model Description

2.1 Summary

We model a dragonfly-like insect with two pairs of wings. The sagittal plane is aligned with the fixed frame (x, z) plane, where z points opposite the direction of gravity. The wings in each pair beat symmetrically about the sagittal plane, and there is zero initial velocity or wind component normal to the sagittal plane, so the motion is constrained to the (x, z) plane, rendering the problem two-dimensional. A subsequent strong simplification is that the body is reduced to a point mass, so there is no coupling between body pitch and wing kinematics. In this manner we can focus entirely on the translational dynamics without concern for flight stability. The dynamical model is nondimensionalized using the body length L, the body mass m, and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the wing kinematics angular parameters, are listed in Table 1. Dimensionless quantities are denoted by an asterisk ($*$).

2.2 Aerodynamic Force Derivation

Each wing is modeled as a single, massless plate element of area A_w with its aerodynamic center at the $\frac{2}{3}$−span station, $r = \frac{2}{3}L_w$, whose lift and drag coefficients are prescribed by functions of the instantaneous angle of attack α. These simplified functions, given by Equation 5 and Equation 6, nevertheless capture the essential phenomena. Parameters for the wing kinematics are defined in Figure 1 and Figure 2 for a single right-hand-side wing. Since the wings of each pair are assumed to beat symmetrically about the sagittal plane, the left-hand-side wing kinematics can be obtained by mirroring about the (x, z) plane.

Each wing element sweeps along an arc of radius $r = \frac{2}{3}L_w$ that lies inside the stroke plane, with normal vector $\vec{n} = \cos\gamma\vec{z} - \sin\gamma\vec{x}$, where γ is the stroke plane inclination angle relative to the horizontal. The sweep

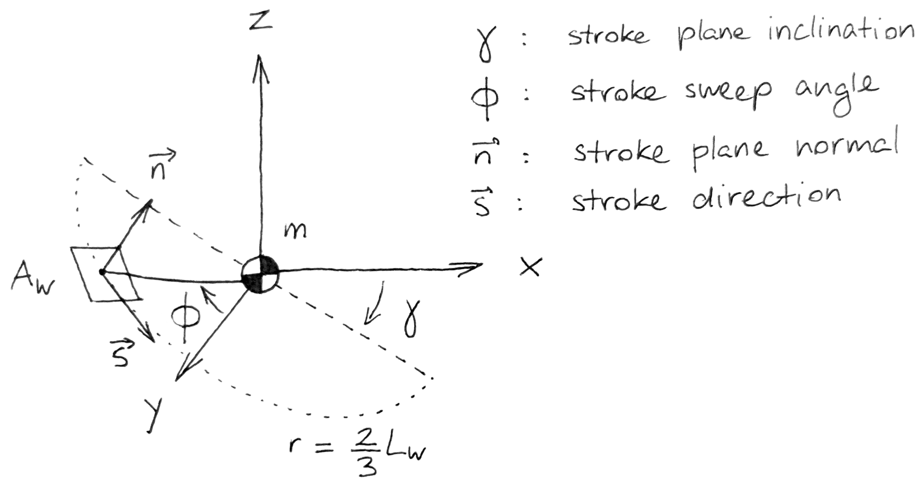

Figure 1: Wing stroke geometry

angle ϕ about $\vec{n}$ is defined relative to the side vector $\vec{y}$, positive counterclockwise about $\vec{n}$. At any point along the wing stroke, the wing velocity relative to the body is

$$\vec{v}_{w/b} = \frac{2}{3} L_w \dot{\phi} \vec{s} \tag{1}$$

with $\vec{s} = \cos\phi\,(\cos\gamma\vec{x} + \sin\gamma\vec{z}) - \sin\phi\vec{y}$. The sweep angle ϕ for wing pair i is prescribed by

$$\phi^{(i)}(t) = \phi_1 \sin\left(\omega t - \sigma_0^{(i)}\right) \tag{2}$$

The phase parameter $\sigma_0^{(i)}$ is set to 0 for the forewing pair, and to σ_0 for the hindwing pair. Both wing pairs beat at the same frequency ω and with the same amplitude ϕ_1. In addition to the sweeping motion, the wing rotates about its own spanwise axis. We refer to this as the wing pitch. It is most easily described in the $(\vec{s}, \vec{n})$ plane. Figure 2 shows the pitch angle ψ defined relative to the stroke plane normal $\vec{n}$. ψ is prescribed for wing pair i by

$$\psi^{(i)}(t) = \psi_0 + \psi_1 \sin\left(\omega t - \delta_0 - \sigma_0^{(i)}\right) \tag{3}$$

Unlike for the sweep angle ϕ, the mean pitch angle ψ_0 may be nonzero, and can be used to redirect the mean aerodynamic force vector, as we will see.

It is convenient to define the chord direction vector $\vec{c} = \cos\psi\vec{n} - \sin\psi\vec{s}$. The angle of attack α is the angle between $\vec{c}$ and $\vec{v}_w$, the projection in the $(\vec{s}, \vec{n})$ plane of the wing velocity relative to the air. Care has to be taken as to the sign of α, as evidenced by the two panels in Figure 2. When $\vec{v}_w \cdot \vec{s} > 0$, α has the correct sign when defined from $\vec{v}_w$ to $\vec{c}$. When $\vec{v}_w \cdot \vec{s} < 0$, it has the correct sign when defined from $\vec{c}$ to $\vec{v}_w$. The following expression gives the correct sign:

$$\alpha = \sin^{-1}\left[\left(\frac{\vec{v}_w}{v_w} \times \vec{c}\right) \cdot (\vec{s} \times \vec{n})\right] \tag{4}$$

Lift and drag coefficients are then obtained by

$$C_L(\alpha) = C_{L,0} \sin 2\alpha \tag{5}$$

$$C_D(\alpha) = C_{D,0} \cos^2\alpha + C_{D,\frac{\pi}{2}} \sin^2\alpha \tag{6}$$

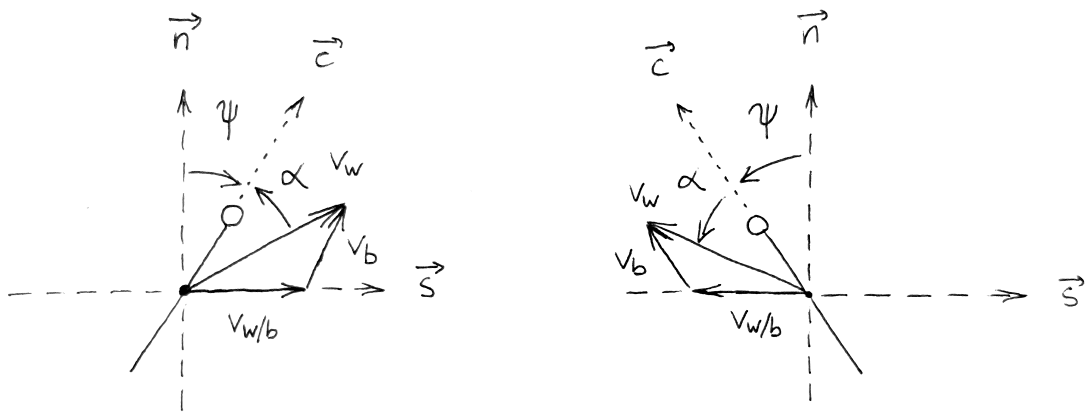

Figure 2: Wing pitch geometry and angle of attack for (a) $\vec{v}_w \cdot \vec{s} > 0$ and (b) $\vec{v}_w \cdot \vec{s} < 0$

with $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, $C_{D,\frac{\pi}{2}} = 2$, as suggested by numerical simulations and experiments across the full range of angle of attack [1, 2]. The lift and drag vectors are, respectively,

$$\vec{d} = -\frac{\vec{v}_w}{v_w} \tag{7}$$

$$\vec{\ell} = \vec{d} \times (\vec{s} \times \vec{n}) \tag{8}$$

And the aerodynamic force of a single wing is

$$\vec{F}_{a,w} = \frac{1}{2}\rho_{\text{air}} A_w v_w^2 \left(C_D \vec{d} + C_L \vec{\ell} \right) \tag{9}$$

2.3 Nondimensionalization

The flying insect is subjected to gravity and the aerodynamic force on its wings. Body drag is neglected. The resulting equation of motion is

$$m\left(\ddot{x}\vec{x} + \ddot{z}\vec{z}\right) = \frac{1}{2}\rho_{\text{air}} A_w \sum_{i}^{N_w} \left[v_w^2 \left(C_D \vec{d} + C_L \vec{\ell} \right) \right]^{(i)} - mg\vec{z} \tag{10}$$

We nondimensionalize it using the body length L, the body mass m, and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the already dimensionless wing angles, are summarized in Table 1. Ranges for the parameters L_w^*, A^*/m^*, and ω^* were obtained from the morphological data in [3] and the wingbeat frequency data in [4] for *Anisoptera*. The resulting dimensionless equation of motion is

$$\ddot{x}^*\vec{x} + \ddot{z}^*\vec{z} = \frac{1}{2}\frac{A^*}{m^*}\left(\frac{2}{3}L_w^*\phi_1\omega^*\right)^2 \frac{1}{N_w}\sum_{i}^{N_w}\left[v_w^{*2}\left(C_D\vec{d} + C_L\vec{\ell} \right) \right]^{(i)} - 1\vec{z} \tag{11}$$

Table 1: Dimensionless parameters

Parameter	Description	Range
$L_w^* = \frac{L_w}{L}$	Wing length ratio	[0.65, 0.85]
$A^*/m^* = \frac{N_w A_w \rho_{\mathrm{air}} L}{m}$	Inverse wing loading	[0.2, 1.0]
$\omega^* = \omega\sqrt{\frac{L}{g}}$	Wing beat frequency	[8, 20]
γ	Wing stroke plane angle	[-180°, 180°]
ϕ_1	Wing flapping amplitude	[0°, 60°]
ψ_0	Mean wing pitch angle	[-90°, 90°]
ψ_1	Wing pitch amplitude	[0°, 90°]
δ_0	Wing pitch phase	[-180°, 180°]
σ_0	Fore/hind phase	[-180°, 180°]

3 Two Analytical Cases

3.1 Stationary Flight

3.1.1 Wingbeat-averaged Aerodynamic Force

In stationary flight, or *hovering*, the wingbeat-averaged aerodynamic force vector is opposed to and equal in magnitude to the weight vector. There are residual oscillations of the body due to fluctuations in the aerodynamic force over a wingbeat cycle, and inertial coupling between the body and wings (not modeled here). Assuming the corresponding body velocities are small compared to the characteristic wing velocity $\frac{2}{3}L_w^*\phi_1\omega^*$, we can approximate the mean aerodynamic force analytically by setting $\dot{x}^* = \dot{z}^* = 0$. As a result, the angle of attack is a simple function of the pitch angle ψ and the instantaneous sign of the flapping velocity:

$$\alpha = -\frac{\dot{\phi}}{|\dot{\phi}|}\left[\frac{\pi}{2} - \psi\right] \tag{12}$$

After making use of the appropriate trigonometric identities, this gives the lift and drag coefficients

$$C_L = -\frac{\dot{\phi}}{|\dot{\phi}|} C_{L,0} \sin 2\psi \tag{13}$$

$$\begin{aligned} C_D &= C_{D,0} \sin^2\psi + C_{D,\frac{\pi}{2}} \cos^2\psi \\ &= \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \cos 2\psi \end{aligned} \tag{14}$$

The aerodynamic force normal to the stroke plane (along $\vec{n}$) is pure lift, and the force within the stroke plane (along $\vec{s}$) is pure drag. Denoting these two components as $F_{w,n}^*$ and $F_{w,s}^*$, respectively, they are

$$F_{w,n}^* = \frac{1}{2}\frac{A_w^*}{m^*}\left(\frac{2}{3}L_w^*\dot{\phi}\right)^2 C_L \tag{15}$$

$$F_{w,s}^* = -\frac{\dot{\phi}}{|\dot{\phi}|}\frac{1}{2}\frac{A_w^*}{m^*}\left(\frac{2}{3}L_w^*\dot{\phi}\right)^2 C_D \tag{16}$$

At this point, it is useful to linearize, in a sense, about the $\phi = 0$ position, by treating $\vec{s}$ as a fixed vector, parallel to the (x, z) plane. This is a common modeling choice, see *e.g.* [5]. The effect of the simplification is that force is increasingly overestimated as the wingbeat amplitude ϕ_1 increases. The effect is small for reasonable values of ϕ_1. Note the numerical model retains the moving $\vec{s}$ (wing beats along an arc, not a straight line) and we only perform the simplification for the analytical derivation, so that the time integral depends on ϕ_1 only as a simple scaling factor. Note further that the simplification has no effect on the mean if the wing pitch is symmetric for both halves of the stroke, since $\langle F_s^* \rangle$ would be zero regardless.

With the above simplification made, we introduce the sweep kinematics $\phi(t^*) = \phi_1 \sin \omega^* t^*$, giving $\dot{\phi} = \phi_1 \omega^* \cos \omega^* t^*$. With the notation $\tau^* = \omega^* t^*$, we evaluate the wingbeat cycle integral $\langle \cdot \rangle = \frac{1}{2\pi} \int_0^{2\pi} (\cdot) d\tau^*$. An important question is how to deal with the sign flips due to the $\dot{\phi}/|\dot{\phi}|$ factors appearing in both F_n^* (through C_L) and F_s^*. We can actually fold them into the "velocity squared" factors as follows

$$
\begin{aligned}
\frac{\dot{\phi}}{|\dot{\phi}|} \dot{\phi}^2 &= \dot{\phi} |\dot{\phi}| \\
&= \phi_1^2 \omega^{*2} \cos \tau^* |\cos \tau^*| \qquad (17)
\end{aligned}
$$

It is also convenient to introduce the wingbeat amplitude shorthand $s_0^* = \frac{2}{3} L_w^* \phi_1$. Evaluating the integrals, and multiplying by the number of wings (A_w^* becomes A^*), we obtain the total cycle-averaged aerodynamic force components

$$
\begin{aligned}
\langle F_n^* \rangle &= \frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \langle C_L |\cos \tau^*|^2 \rangle \\
&= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 C_{L,0} \langle \sin 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \qquad (18) \\
\langle F_s^* \rangle &= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \langle C_D \cos \tau^* |\cos \tau^*| \rangle \\
&= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \qquad (19)
\end{aligned}
$$

Note the constant drag term vanishes because $\langle \cos \tau^* |\cos \tau^*| \rangle = 0$. The expressions feature a common $\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2$ factor, and a pitch- and aerodynamic-coefficient-dependent factor, which is lift-based for $\langle F_n^* \rangle$, and drag-based for $\langle F_s^* \rangle$. At this point, the only kinematics assumption we have made is sinusoidal flapping of the wings, while ψ remains generic. First, we can recover the analytical result of [5] for symmetric flapping with $\psi = -\frac{\pi}{4}$ for the $+\vec{s}$ stroke, and $\psi = +\frac{\pi}{4}$ for the $-\vec{s}$ stroke. Noting that $\cos \tau^* > 0$ for the $+\vec{s}$ stroke and < 0 for the $-\vec{s}$ stroke, and that $\langle \cos^2 \tau^* \rangle = \frac{1}{2}$, we recover the result

$$
\begin{aligned}
\langle F_n^* \rangle &= \frac{1}{4} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 C_{L,0} \qquad (20) \\
\langle F_s^* \rangle &= 0 \qquad (21)
\end{aligned}
$$

We could obtain this result with our numerical model if we were to replace the sine wave with a square wave, and set $\psi_0 = 0$, $\psi_1 = \frac{\pi}{4}$. In reality, dragonfly wing pitch kinematics follow something between a sine wave and a square wave (*i.e.* a somewhat flattened sine wave). We stick to a sine wave here so as not to introduce higher harmonics ([6]) or a "flattening" coefficient ([7]), while keeping the more physical non-instantaneous stroke reversal (would be instantaneous with a square wave). To look at the effect of the pitch parameters on the force magnitude in the stroke plane and stroke-plane-normal directions, we introduce

$$
\begin{aligned}
q^* &= \frac{1}{4} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \qquad (22) \\
C_{F^*,n} &= -2 C_{L,0} \langle \sin 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \qquad (23) \\
C_{F^*,s} &= -\left(C_{D,\frac{\pi}{2}} - C_{D,0} \right) \langle \cos 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \qquad (24)
\end{aligned}
$$

such that $\langle F_n^* \rangle = C_{F^*,n} q^*$ and $\langle F_s^* \rangle = C_{F^*,s} q^*$. It is also convenient to introduce the overall magnitude $C_{F^*} = \sqrt{C_{F^*,n}^2 + C_{F^*,s}^2}$.

3.1.2 Power Efficiency

In addition to the mean aerodynamic force, another quantity of interest is the mean power developed over a wingbeat. Because we assume the wings are massless, power is only due to the work of the aerodynamic force. In the hovering case, only the drag force does work, because the lift force is strictly normal to $\vec{s}$. It follows the (dimensionless) wingbeat-averaged power is

$$\begin{aligned}\langle P^* \rangle &= \langle |F_s^* s_0^* \omega^* \cos \tau^*| \rangle \\ &= \frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^*\right)^3 \langle C_D |\cos \tau^*|^3 \rangle \end{aligned} \tag{25}$$

Following the derivation of the *cost of endurance* in [2], we can normalize the above by introducing the quantity $u_{\text{ref}} = \sqrt{2mg/\rho A}$, more specifically its dimensionless form $u^*_{\text{ref}} = \sqrt{2/\frac{A^*}{m^*}}$, giving

$$\begin{aligned}\frac{\langle P^* \rangle}{u^*_{\text{ref}}} &= \frac{1}{2\sqrt{2}} \left[\frac{A^*}{m^*} \left(s_0^* \omega^*\right)^2 \right]^{3/2} \langle C_D |\cos \tau^*|^3 \rangle \\ &= 2\sqrt{2} \left(q^*\right)^{3/2} \langle C_D |\cos \tau^*|^3 \rangle \end{aligned} \tag{26}$$

When hovering, the wingbeat-average net aerodynamic force $C_{F^*} q^* = 1 \implies q^* = 1/C_{F^*}$. We introduce the hovering power coefficient

$$C_{P^*} = 2\sqrt{2} \frac{\langle C_D |\cos \tau^*|^3 \rangle}{C_{F^*}^{3/2}} \tag{27}$$

which is analogous to the steady value $C_D/C_L^{3/2}$. The velocity-weighted drag coefficient in the numerator plays the role of C_D, and the force coefficient C_{F^*} in the denominator, which is, in general, part lift and part drag, plays the role of C_L. To be more specific, C_{F^*} is lift-only as long as the wing stroke is symmetric ($\psi_0 = 0$), which, for hovering, requires the stroke plane to be horizontal. Hovering at an inclined stroke plane requires, as we will see in subsubsection 3.1.3, $\psi_0 \neq 0$, resulting in drag contributing to C_{F^*}.

3.1.3 Force-Maximizing and Power-Minimizing Hover Kinematics

$C_L(\alpha)$ and $C_D(\alpha)$ being fixed, the force and power coefficients C_{F^*} and C_{P^*} defined in subsubsection 3.1.1 and subsubsection 3.1.2 respectively, depend only on the pitch parameters ψ_0, ψ_1, and δ_0. We visualize these coefficients across two 2D slices, one at $\psi_0 = 0°$, and one at $\delta_0 = 90°$, in Figure 3 and Figure 4. The optimal values of each coefficient (maximum C_{F^*} and minimum C_{P^*}), indicated with white dots, both occur at $\psi_0 = 0°$ and $\delta_0 = 90°$, with different values of the pitching amplitude ψ_1. Unsurprisingly, the C_{F^*} maximum occurs close to $\psi_1 = 45°$, which maximizes lift, while the C_{P^*} minimum occurs close to $\psi_1 = 90°$, which minimizes drag. To be exact, there are also maxima/minima at $\psi_0 = \pm 90°$. They are interesting, but do not seem to correspond to typical dragonfly wingbeat patterns, so we disregard them here.

An important consequence of the maximum and minimum occuring at $\psi_0 = 0°$ is that energy-efficient wing pitch occurs for a symmetric wingbeat, which results in a net mean force normal to the stroke plane, necessitating the use of a horizontal stroke plane. In other words, our quasi-steady lift and drag model is unable to account for the efficiency advantage of the dragonfly inclined stroke plane evidenced in other work [1, 2]. Since we want the model kinematics to still resemble dragonfly flight, we will impose a nonzero hover stroke plane angle γ (see section 4), independent of efficiency considerations for the present model. This will require the wingbeat-averaged force vector to be angled relative to the stroke plane normal by an amount $-\gamma$. As we will now see, this can be achieved with nonzero ψ_0. Let us denote the angle of the force

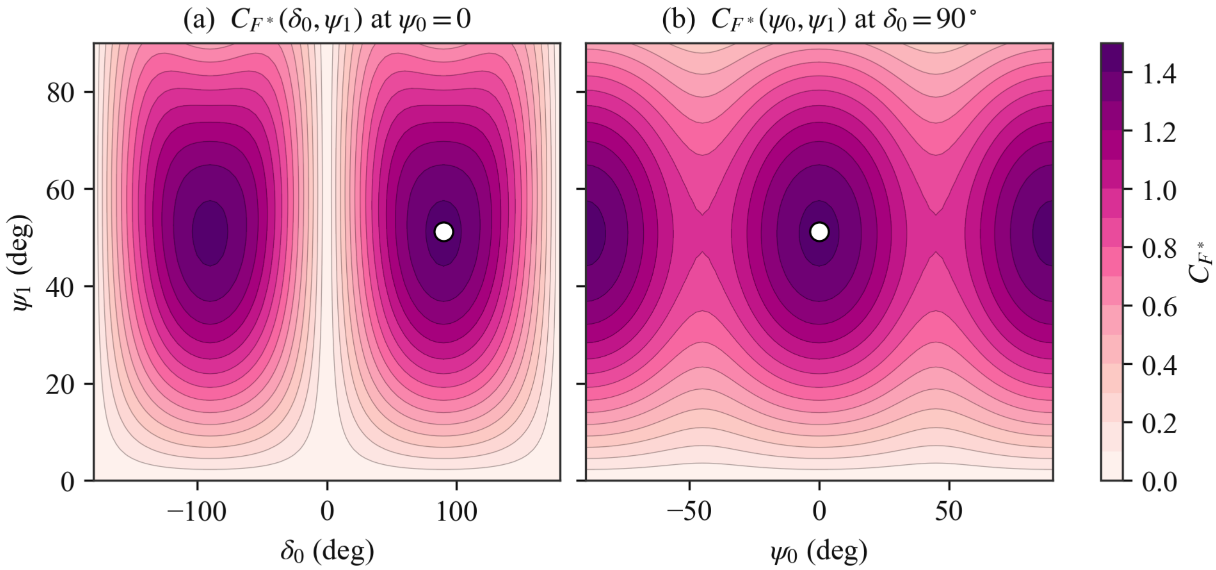

Figure 3: Dependence of C_{F^*} on pitch parameters at zero body velocity. Maximum shown as white dots.

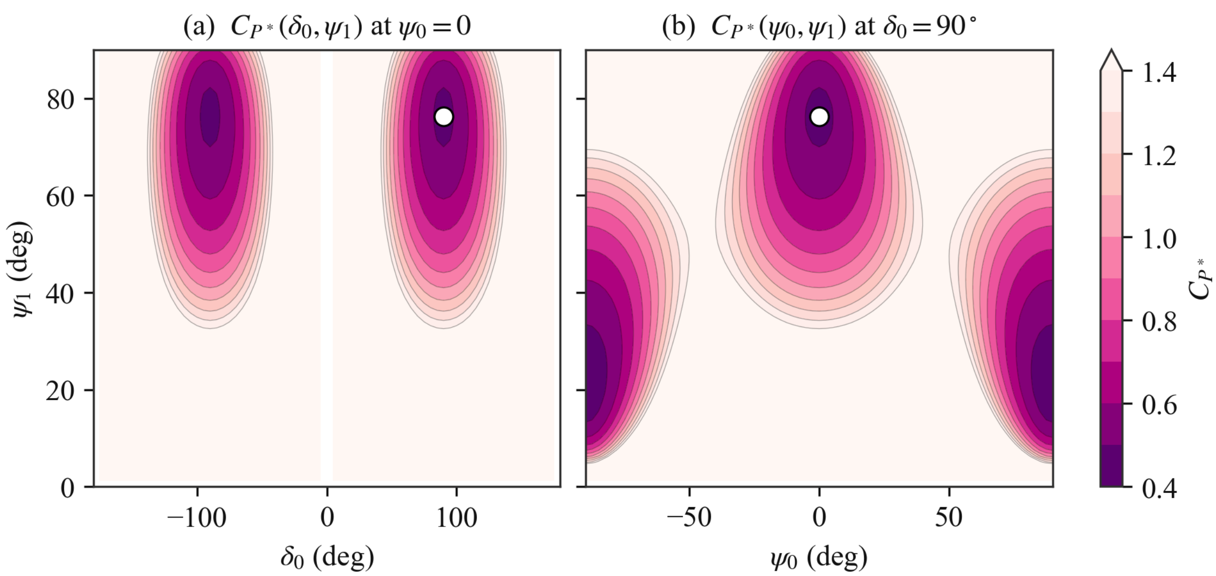

Figure 4: Dependence of C_{P^*} on pitch parameters for hovering. Minimum shown as white dots.

vector to the stroke plane normal $\vec{n}$ as β. Since the wingbeat-averaged force is $\langle \vec{F}^* \rangle = q^* \left(C_{F^*,n}\vec{n} + C_{F^*,s}\vec{s} \right)$, β follows from the $C_{F^*,n}$ and $C_{F^*,s}$ coefficients (Equation 23 and Equation 24) as

$$\beta = -\,\mathrm{atan2}\left(C_{F^*,s},\ C_{F^*,n} \right), \tag{28}$$

taken positive counterclockwise from $\vec{n}$ (viewed with $\vec{x}$ pointing right and $\vec{z}$ up). Thus $\beta = 0$ for a force along $\vec{n}$, and $\beta = -\gamma$ aims the mean force along the vertical for hover at stroke plane angle γ. Figure 5 maps the two components over the pitch plane. $C_{F^*,n}$ is largest for a near-symmetric stroke, whereas $C_{F^*,s}$ vanishes at $\psi_0 = 0$ and grows with $|\psi_0|$, supplying the off-normal force direction that an inclined stroke plane hover requires.

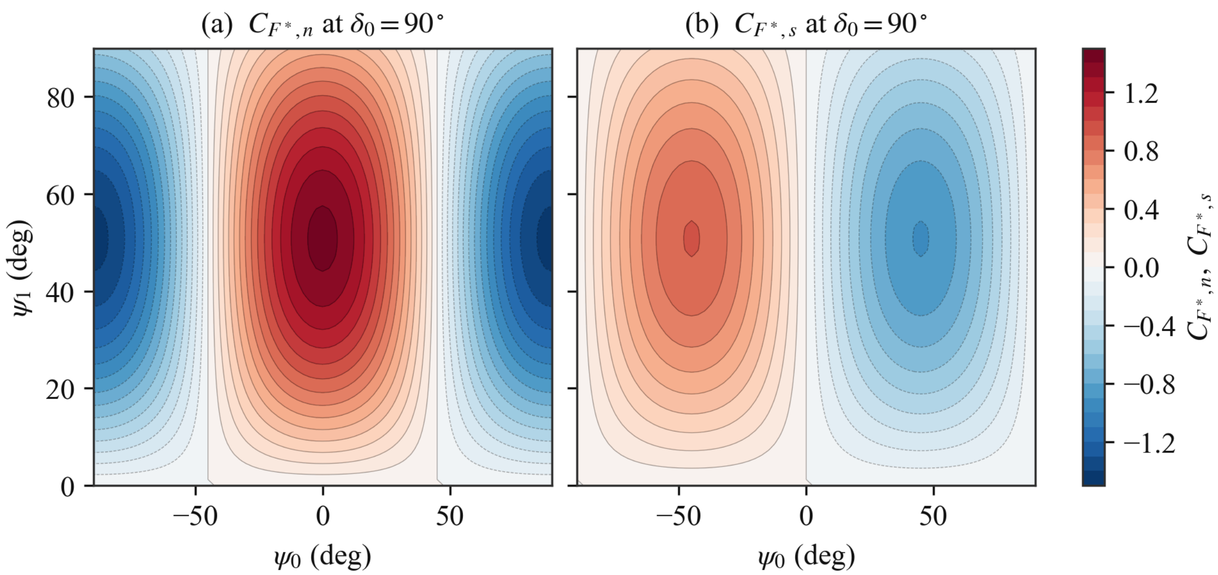

Figure 5: Stroke-plane-normal (lift) component $C_{F^*,n}$ (a) and in-plane (drag) component $C_{F^*,s}$ (b) of the force coefficient over the pitch plane at $\delta_0 = 90^\circ$. The force direction quadrant in the $(\vec{s}, \vec{n})$ plane can be inferred from the signs.

We further show that β depends on ψ_0 alone, independent of the pitch amplitude ψ_1 and the phase magnitude $|\delta_0|$. The weight $\cos\tau^*|\cos\tau^*|$ in Equation 23 and Equation 24 is antisymmetric under $\tau^* \to \tau^* + \pi$, a shift that also maps $\psi \to 2\psi_0 - \psi$ (since $\sin(\tau^* + \pi - \delta_0) = -\sin(\tau^* - \delta_0)$). Splitting each cycle average over the two half-strokes and applying the sum-to-product identities gives

$$\langle \sin 2\psi\ \cos\tau^*|\cos\tau^*| \rangle = \quad \cos 2\psi_0\ K, \tag{29}$$

$$\langle \cos 2\psi\ \cos\tau^*|\cos\tau^*| \rangle = -\sin 2\psi_0\ K, \tag{30}$$

where the factor

$$K = \frac{1}{\pi}\int_0^{\pi} \sin[2\psi_1 \sin(\tau^* - \delta_0)]\cos\tau^*|\cos\tau^*|\, d\tau^* \tag{31}$$

contains the entire ψ_1 and $|\delta_0|$ dependence. It enters $C_{F^*,n}$ and $C_{F^*,s}$ as a common factor and so cancels in their ratio. Only its sign remains, fixed by the sign of δ_0, flipping the force by 180°. With Equation 28 this yields the closed form

$$\beta = \mathrm{atan2}\Big[\Big(C_{D,\frac{\pi}{2}} - C_{D,0} \Big) \sin 2\psi_0,\ 2C_{L,0}\cos 2\psi_0 \Big] \tag{32}$$

for $\delta_0 > 0$. In particular $\beta = 0$, $90°$, $180°$ at $\psi_0 = 0$, $45°$, $90°$, and near hover $\beta \approx \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{C_{L,0}} \psi_0 \approx 1.3\,\psi_0$. The full numerical model retains the moving stroke tangent ($\vec{s}$ not constant) and departs from Equation 32 by at most $\sim 1°$ over the ψ_1 and $|\delta_0|$ ranges, so the independence is exact in the analytical model and very nearly so in practice. We show $\beta(\psi_0)$ in Figure 6 along with the maximum C_{F^*} at a given ψ_0.

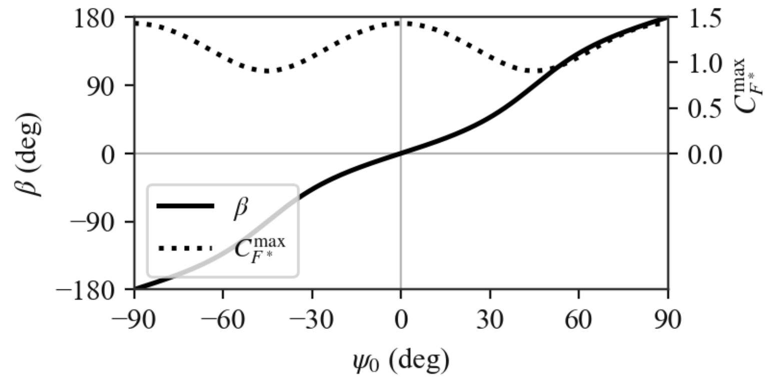

Figure 6: Effect of ψ_0 on force direction and wing stroke efficiency.

3.1.4 Implications for Morphology and Wingbeat Frequency

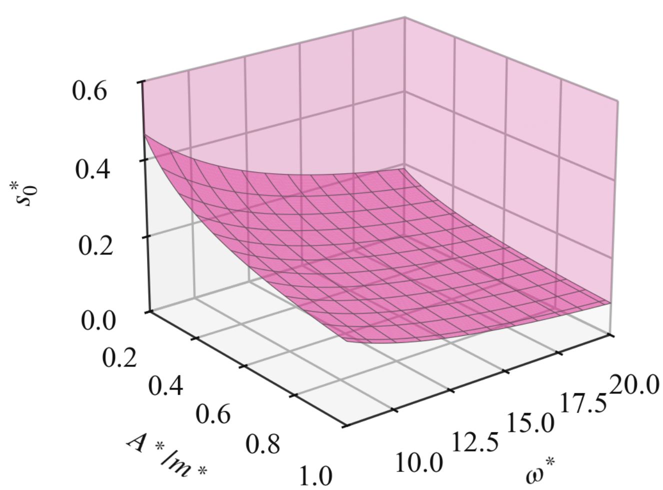

Figure 7: Approximate region of hover feasibility, with boundary surface.

Now that we have identified two families of “optimal” wing pitch kinematics, independent of morphology and wingbeat frequency, we take a brief look at what our findings imply about these parameters. Using the hovering criterion $F^* = 1$ for the maximum C_{F^*} pitch parameters, Figure 7 shows the required wingbeat amplitude s_0^* (function of L_w^* and ϕ_1) for a given $\left(\frac{A^*}{m^*}, \omega^*\right)$. The sweep amplitude ϕ_1 can be recovered by

$$\phi_1 = \frac{3}{2}\frac{s_0^*}{L_w^*}$$

The very high amplitudes ($\phi_1 \approx 54°$) associated with low values of both $\frac{A^*}{m^*}$ and ω^* suggest such combinations probably do not occur, and that the parameter space can be further restricted. The data source for ω^* does not report measurements necessary to obtain $\frac{A^*}{m^*}$, but we could maybe find other sources. We can estimate the typical hover wingbeat amplitude by assuming that the lowest ω^* occur at the highest $\frac{A^*}{m^*}$. For this set of parameters ($\omega^* = 8$ and $\frac{A^*}{m^*} = 1.0$) the amplitude is $s_0^* \approx 0.21$, which corresponds to $\phi_1 \approx 24°$.

For the rest of the report, we use a reference configuration with intermediate values of all three parameters, yielding hover amplitudes close to the above estimate. The parameter values are shown in Table 2.

Table 2: Reference configuration

A^*/m^*	L_w^*	ω^*	γ
0.6	0.75	14	−40°

3.2 Stroke-Plane-Normal Flight

3.2.1 Rationale

Dragonflies have been observed to fly with their stroke plane normal $\vec{n}$ approximately aligned to the direction of travel [6]. We treat that aligned configuration here, with the body advancing along $\vec{n}$, and use the analysis for controller design in subsection 4.2.

3.2.2 Wingbeat-averaged Aerodynamic Force

We now add a velocity component normal to the stroke plane, v_n^*. The wing then meets the air at an angle $\theta = \operatorname{atan2}(v_s^*, v_n^*)$ from the stroke plane normal $\vec{n}$, and the angle of attack becomes

$$\alpha = \psi + \theta, \qquad \theta = \operatorname{atan2}(v_s^*, v_n^*). \tag{33}$$

Note that as the normal velocity vanishes ($v_n^* \to 0$), θ returns to $\pm\frac{\pi}{2}$, and we recover the hover angle of attack of subsubsection 3.1.1. We parametrize v_n^* through the advance ratio

$$J = \frac{v_n^*}{s_0^*\omega^*}, \tag{34}$$

We have $v_s^* = s_0^*\omega^* \cos\tau^*$ and $v_n^* = J s_0^*\omega^*$, so the wing velocity relative to the air is $v_w^* = s_0^*\omega^*\sqrt{\cos^2\tau^* + J^2}$. We substitute these into the lift and drag of Equation 5 and Equation 6, and cycle-average in the small-ϕ limit, freezing the stroke basis as in subsubsection 3.1.1. This gives the moving-flight coefficient

$$C_{F^*}(\psi_0, \psi_1, \delta_0; J) = \sqrt{C_{F^*,n}^2 + C_{F^*,s}^2}. \tag{35}$$

Unlike in hover, the normal velocity tilts the relative wind off the stroke plane, so lift and drag each project onto both stroke axes. Writing $C_L = C_{L,0}\sin 2\alpha$ and $C_D = C_{D,0}\cos^2\alpha + C_{D,\frac{\pi}{2}}\sin^2\alpha$ with α from Equation 33, the stroke-plane-normal and in-plane coefficients are

$$C_{F^*,n} = \ \ 2\left\langle \sqrt{\cos^2\tau^* + J^2}\ \left(C_L\cos\tau^* - J\,C_D\right)\right\rangle, \tag{36}$$

$$C_{F^*,s} = -2\left\langle \sqrt{\cos^2\tau^* + J^2}\ \left(C_D\cos\tau^* + J\,C_L\right)\right\rangle. \tag{37}$$

These reduce to the pure-lift Equation 23 and pure-drag Equation 24 of subsubsection 3.1.1 as $J \to 0$. The cross terms $J\,C_D$ and $J\,C_L$ are the new contributions from the normal velocity, absent in hover. Note that the entire velocity dependence has collapsed onto the single parameter J.

The same parity argument as in subsubsection 3.1.1 applies here. For symmetric pitch ($\psi_0 = 0$, $\delta_0 = \frac{\pi}{2}$), the wing pitch is $\psi = -\psi_1\cos\tau^*$. The substitution $\tau^* \to \pi - \tau^*$ sends $v_s^* \to -v_s^*$ at fixed v_n^*, hence $\theta \to -\theta$ and $\psi \to -\psi$, so $\alpha \to -\alpha$. Under this reflection C_L is odd and C_D even, so the $C_{F^*,s}$ integrand is odd and the $C_{F^*,n}$ integrand even. The in-plane component therefore vanishes, $C_{F^*,s} = 0$, and the wingbeat-averaged force lies exactly along the stroke plane normal at every advance ratio. As in hover, the aligned wingbeat produces a force purely along the flight axis, and redirecting it away from $\vec{n}$ requires breaking the symmetry through a nonzero mean pitch ψ_0. With $C_{F^*,s} = 0$ the magnitude reduces to the single integral

$$C_{F^*}(J) = 2\left|\left\langle \sqrt{\cos^2\tau^* + J^2}\ \left(C_L\cos\tau^* - J\,C_D\right)\right\rangle\right|. \tag{38}$$

This returns to the hover value as $J \to 0$ and decreases monotonically with J, as the steady normal velocity detunes the angle of attack away from its lift-producing range (Figure 8(a)).

The pitch amplitude that maximizes C_{F^*} also retunes with the normal velocity. The optimum $\psi_1^{\max}(J)$ decreases from its hover value as J grows. The wing pitches less, because the steady normal velocity now sets part of the angle of attack that the sweep set on its own in hover. This partly recovers the force lost at fixed pitch (Figure 8(b)). We track the lift-producing optimum here. There is also a second branch at larger ψ_1, where the wing is carried backward by the normal velocity and the wingbeat-averaged force opposes $\vec{n}$. This branch cannot support weight, so we discard it. The schedule $\psi_1^{\max}(J)$ is the one the generalized controller of subsection 4.2 slaves to.

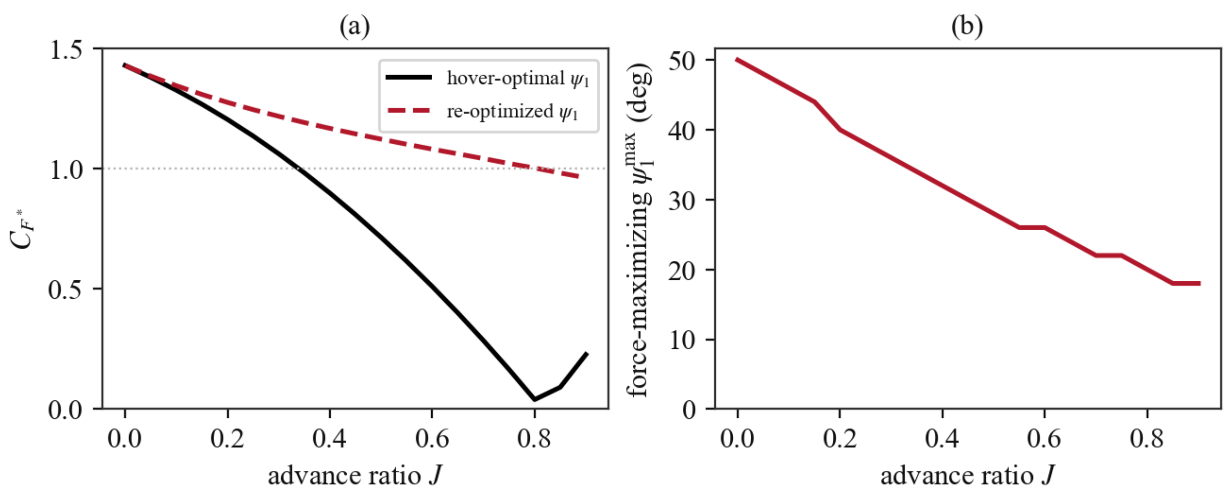

Figure 8: Axial flow, reference configuration. (a) Force coefficient $C_{F^*}(J)$ at the fixed hover-optimal pitch and at the re-optimized (lift-branch) pitch. (b) The force-maximizing amplitude $\psi_1^{\max}(J)$.

4 Controller Design

4.1 Hover Controller

4.1.1 Design Principle

Insights from subsection 3.1 suggest a few possible ways to achieve hover control. Candidates for control variables can be split into two groups: those that control force magnitude (ϕ_1, ω^* through q^* and ψ_1, δ_0 through C_{F^*}), and those that control force direction (γ, ψ_0). Note this is not such a clean split, because ψ_0 also changes force magnitude through C_{F^*}, just more weakly than it changes direction. For simplicity, we would rather consider ω^* to be a fixed parameter, so we will exclude it at least for now. Since we also do not fully understand the factors that set the hovering stroke plane angle, we will also consider γ to be fixed for now (later on, we compare the effectiveness of γ and ψ_0 as directional control variables). In addition, it is beneficial to keep C_{F^*} at or near its maximum value, which sets ψ_1 and δ_0. This leaves ϕ_1 (or s_0^*) for magnitude and ψ_0 for direction. At equilibrium, ψ_0 will be such that $\beta = -\gamma$ (force is vertical), and ϕ_1 such that $F_a^* = 1$. Table 3 shows the required value of ψ_0 for different γ, for a reference configuration $L_w^* = 0.75$, $\frac{A^*}{m^*} = 0.6$, and $\omega^* = 14$. Different values of these parameters impose a different ϕ_1 for hover, which results in small changes in ψ_0 up to a few percent.

Table 3: Mean pitch for hover at various stroke plane angles for reference configuration

γ	ψ_0	$C_{F^*}^{\max}$
0°	0°	1.43
−10°	7.9°	1.40
−20°	15.1°	1.31
−30°	21.4°	1.21
−40°	26.7°	1.11

ϕ_1 is set by

$$\phi_1^{\text{hover}} = 3\left(L_w^* \omega^* \sqrt{C_{F^*}^{\max}(\psi_0)\frac{A^*}{m^*}}\right)^{-1}$$

4.1.2 Departure from Wingbeat-Averaged Picture

Before designing a controller, we look at how the actual, unsteady hover dynamics differ from the wingbeat-averaged picture we have been working with so far. Indeed the body velocity will not be exactly zero throughout, and will oscillate according to the fluctuations of the aerodynamic force over a wingbeat cycle. There is an additional effect due to small but nonzero mass of the wings, which will make the body bob up and down in the direction of the flapping motion. We have neglected this in our work so far. In this part we will evaluate the relative importance of the two effects on the body oscillations.

A key parameter here is the fore/hind wing phase shift σ_0. We expect the oscillations to be more pronounced when the wings beat in-phase ($\sigma_0 = 0°$) than when they beat out-of-phase ($\sigma_0 = 180°$). In fact, if the forewing and hindwing motions are identical except for the phase shift, the wing inertia effect will be zero (true for translation, not for body pitch, which we do not consider in the present model).

We introduce a relative wing mass parameter $m_w^* = m_w/m$. According to [3], values range from 0.01 to 0.03 for *Anisoptera*. We run one hover wingbeat to obtain the body oscillation peak-to-peak amplitude, comparing the ideal massless wing case to the $m_w^* = 0.02$ case, for the reference configuration of Table 2 (Figure 9). The wing mass and aerodynamic contributions to the body oscillations appear to be of the same order, except, as expected, when the wings beat out-of-phase. We will consider $m_w^* = 0$ in the following, but should keep in mind that this is only effectively true at $\sigma_0 = 180°$.

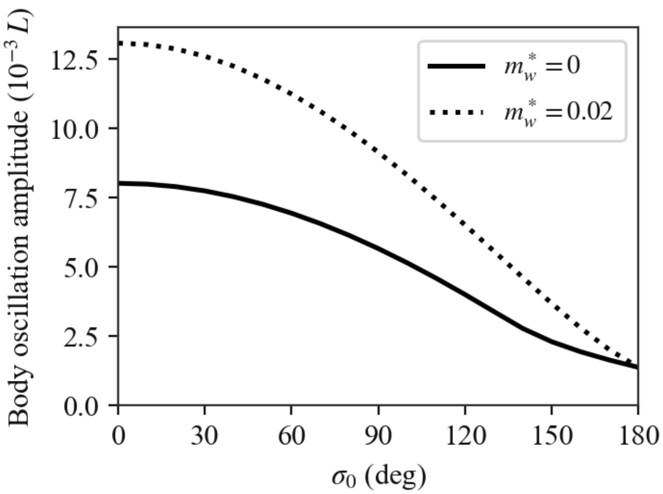

Figure 9: Effect of σ_0 and m_w^* on body oscillations during hover.

Another effect of the aerodynamic force oscillations is positional drift. For a constant, open-loop wingbeat pattern, the dragonfly settles in a steady-state that has a small but nonzero mean body velocity, when the equilibrium prediction is that the zero mean force condition occurs at exactly zero mean body velocity.

Overall these effects are relatively mild, with instantaneous body velocities of approximately $0.25 s_0^* \omega^*$ in the worst case scenario ($\sigma_0 = 0°$, $\omega^* = 8$, $A^*/m^* = 1.0$), and only about $\frac{1}{10}$ of that for $\sigma_0 = 180°$, so the conclusions drawn from the equilibrium analysis should be applicable to controller design.

4.1.3 Proposed Control Scheme

Recall we identified ϕ_1 and ψ_0 as the two control variables for hover. ϕ_1 controls force magnitude, and ψ_0 controls force direction, with the secondary effect of reducing the force magnitude as it moves away from the stroke plane normal mean $\psi_0 = 0$. It seems natural to use ϕ_1 to control altitude, and ψ_0 to control fore-aft drift by tilting the force direction vector from the vertical. Specifically we propose the following control scheme

$$\psi_0 = \psi_0^{\text{eq}} - K_x(x - x_0)$$
$$\phi_1 = \phi_1^{\text{eq}} - K_z(z - z_0)$$

Note that orienting the force in the $-x$ direction requires pitching up (increasing ψ_0), so we will need $K_x < 0$. To design the proportional gains for the reference configuration (Table 2) we begin by forming the Jacobian matrix of the wingbeat-averaged aerodynamic force $\vec{F}_a^*$ at the hover equilibrium, using finite differences, to evaluate the control effectiveness of the two variables about the force axes. The Jacobian is

$$\mathbf{J} = \begin{bmatrix} \dfrac{\partial F_{a,x}^*}{\partial \psi_0} & \dfrac{\partial F_{a,x}^*}{\partial \phi_1} \\ \dfrac{\partial F_{a,z}^*}{\partial \psi_0} & \dfrac{\partial F_{a,z}^*}{\partial \phi_1} \end{bmatrix} \approx \begin{bmatrix} -2.06 & -0.05 \\ +0.93 & 5.83 \end{bmatrix}$$

Notably ϕ_1 has a negligible effect on force tilt from the vertical ($F_{a,x}^*$), while as we pointed out before ψ_0 does have a significant secondary effect on $F_{a,z}^*$. Since ϕ_1 cleanly drives $F_{a,z}^*$, its control loop will absorb the

ψ_0 cross term. The decoupled dynamics are (note mass matrix is I because of nondimensionalization)

$$\ddot{x}^* = -2.06 \times \delta\psi_0$$
$$\ddot{z}^* = 5.83 \times \delta\phi_1$$

So $\omega_n^2 = -2.06K_x = 5.83K_z$. We want ω_n slow enough that the controller sees the wingbeat-averaged force and not the fluctuations. Using $\omega_n = 1.5$ (10 times smaller than the reference wingbeat frequency $\omega^* = 14$) we get

$$K_x = -1.09$$
$$K_z = 0.39$$

4.1.4 Results

The step response is shown in Figure 10. The figure shows the reference configuration along with low and high wing loading configuration. A^*/m^* was chosen as the variable here because it had the greatest effect on the damping of the response. The response is very noticeably more damped as A^*/m^* increases. Over the range of parameters, using the same gains designed for the reference configuration, the achieved ω_n was close to the design value of 1.5, and the response did not vary very much except along the A^*/m^* axis. The controller was stable across the full parameter space.

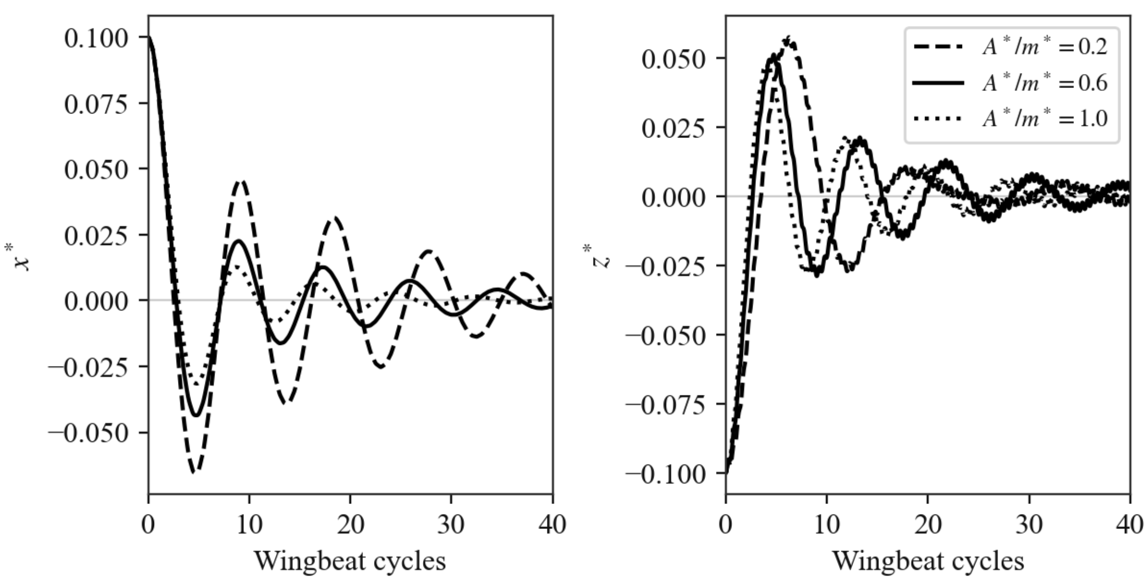

Figure 10: Effect of A^*/m^* on hover controller step response. Each panel is driven by an independent single-axis displacement (the x^* panel by an x offset, the z^* panel by a z offset) so that each curve reflects that axis's own response, isolated from the cross-axis coupling.

A more interesting result is the influence of stroke plane angle on damping, as shown in Figure 11. It appears that damping is strongest within the stroke plane, which means that vertical damping in the $\gamma = 0°$ case is relatively low, and increases as the stroke plane is tilted. Horizontal damping decreases accordingly, with the end result being that there is a better match in the damping characteristics along both axes. Perhaps this is another advantage of hovering with an inclined stroke plane.

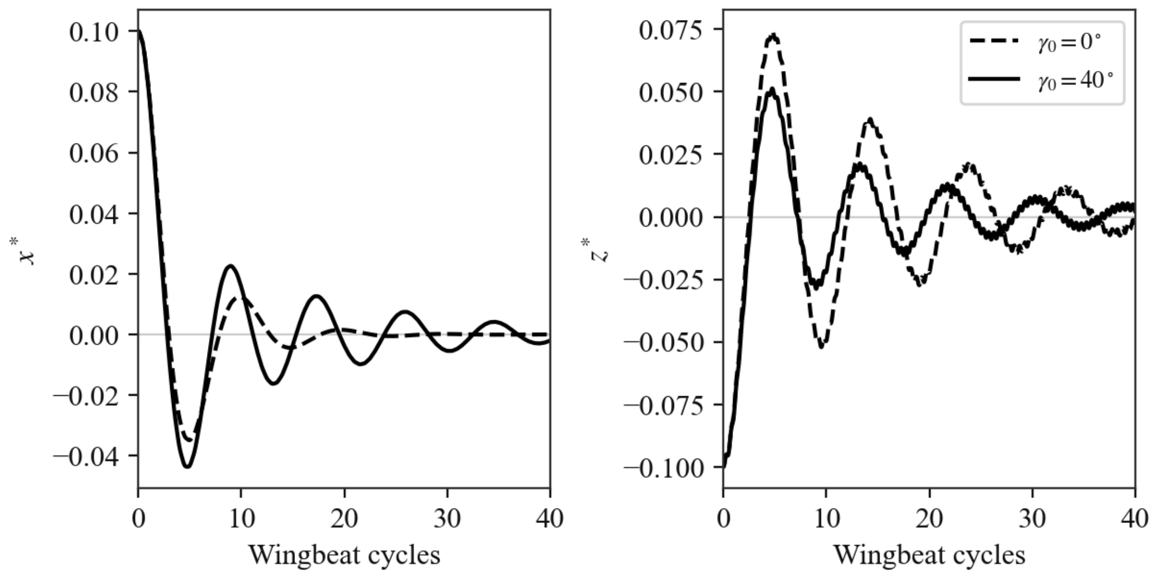

Figure 11: Effect of γ on hover controller step response. Each panel is driven by an independent single-axis displacement (the x^* panel by an x offset, the z^* panel by a z offset) so that each curve reflects that axis's own passive damping, isolated from the cross-axis coupling. Gains are retuned per case to a common ω_n, leaving the damping ratio as the only difference.

4.2 Generalized Controller

4.2.1 Design Principle

Proposal to extend hover controller to moving/maneuvering flight

- Similar inner loop, with ϕ_1 controlling magnitude and ψ_0 controlling direction
- δ_0 still fixed at 90°
- ψ_1 becomes slaved to advance ratio, always giving maximum C_{F^*}
- γ becomes slaved to flight direction and advance ratio. At zero advance ratio, it is set to the hover value (fixed parameter, $\gamma_h = -40°$). In the high limit, it is set to γ_f such that $\vec{n}$ points in the body velocity direction. In between, γ is set to $f(J)\gamma_f + (1 - f(J))\gamma_h$, with $f(J)$ between 0 and 1, $f'(J) \geq 0$, $f(0) = 0$, $f(\infty) = 1$.
- Because $\gamma_h \neq 0$, there is an asymmetry in this basic formulation between flight along $+x$ and along $-x$. We can resolve this by flipping the sign of γ_h depending on the desired direction of motion.

The controller should take into account the inclined hover stroke plane, and transition smoothly between and hoving and steady motion. The in-plane body velocity component should remain small compared to the wing velocity relative to the body.

4.2.2 Proposed Control Scheme

We realize the principle above as a velocity-scheduled controller. The pitch phase is held fixed at $\delta_0 = 90°$. The stroke plane angle is blended between its hover and aligned values as

$$\gamma(J) = f(J)\,\gamma_f + \big(1 - f(J)\big)\,\gamma_h, \qquad f(J) = 1 - e^{-(J/J_c)^2}. \tag{39}$$

The aligned value $\gamma_f = \operatorname{atan2}(-v_x^*, v_z^*)$ points the stroke plane normal $\vec{n}$ along the body velocity (axial flow). The hover value $\gamma_h = -40°\ \operatorname{sign}(v_x^*)$ is the inclined hover plane of Table 2, with its sign chosen so that it leans toward the commanded direction. The blend f is monotone, with $f(0) = 0$ and $f(\infty) = 1$ (we use $J_c = 0.35$). The pitch amplitude ψ_1 is slaved to the advance ratio, taking the force-maximizing value $\psi_1^{\max}(J)$ of subsubsection 3.2.2 (Figure 8). This leaves the two hover handles, ϕ_1 for magnitude and ψ_0 for direction. We trim them to the required force by the same 2×2 inversion as the hover inner loop, now evaluated at the current body velocity. Body drag is neglected, so steady straight flight requires the weight-supporting force at every speed. This schedule is therefore the equilibrium trim, and maneuvering is the transient excursion of (ϕ_1, ψ_0) about it.

4.2.3 Results

We trim the controller to the weight-supporting force along forward, climb, and descent velocity rays (Figure 12). In forward flight, the schedule tilts the stroke plane toward the vertical, so the wing meets the air along its normal. The wingbeat-averaged force then stays near vertical, with little redirection needed. The mean pitch ψ_0 decreases with speed and stays well clear of its bound, the flapping amplitude ϕ_1 stays small, and the aerodynamic power is nearly flat (Figure 12(a–c)).

In vertical flight, the stroke plane instead keeps its inclined hover value at low-speed climb, and relaxes toward the aligned orientation as the normal velocity grows (Figure 12(d–f)). Forward flight and climb both remain feasible beyond $J = 1$. Descent is the limiting case. At the minimum flapping amplitude, the normal velocity already supplies more than the weight, which restricts the trim to $J \approx 0.35$.

We now run the controller under the unsteady dynamics, with the control updated once per wingbeat. It tracks the commanded body velocity in two maneuvers (Figure 13). A forward dash follows an accelerate–cruise–decelerate command to $J \approx 0.5$. A pull-up follows the velocity command through a quarter-turn from forward flight to climb.

A more demanding test reverses the body velocity in both axes. We command the controller around a closed circular path (forward, climb, backward, descent, and back to rest), so that v_x^* and v_z^* each change sign twice (Figure 14). The loop must be kept gentle ($J \lesssim 0.16$). A faster loop fails at the bottom of the circle, where the body descends as v_x^* reverses. There the axial descent geometry and the $\pm x$ stroke plane sign flip coincide, ψ_0 saturates at its bound, and the body drops out of the loop (the same descent limit found above).

The same velocity tracker also drives a pure-pursuit prey capture. At each control update, we aim the commanded velocity along the line of sight to the prey, at a fixed speed exceeding the prey's. Starting from rest, the dragonfly closes the range monotonically and intercepts a horizontally moving target (Figure 15). A descending target would instead inherit the descent limit noted above.

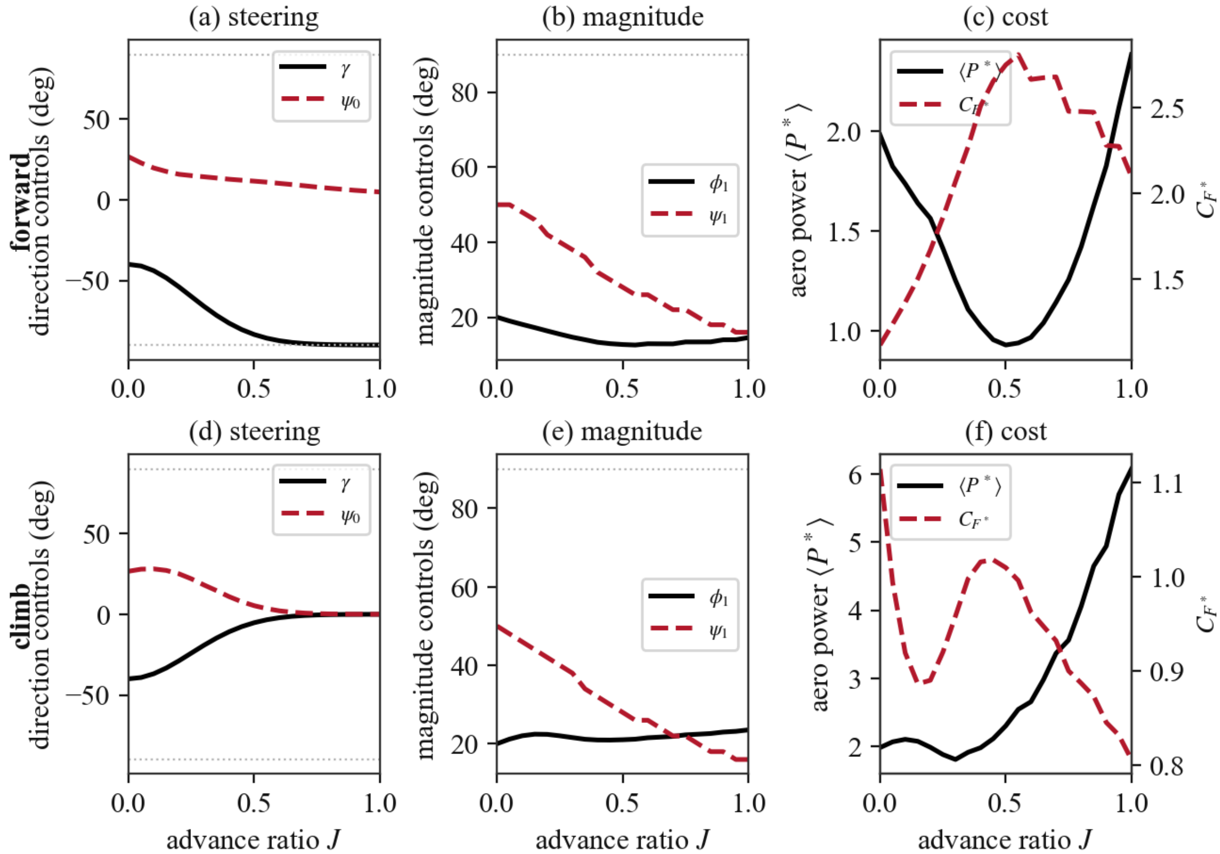

Figure 12: Generalized controller, reference configuration. Top row forward flight, bottom row climb. (a,d) Direction controls γ and ψ_0. (b,e) Magnitude controls ϕ_1 and ψ_1. (c,f) Aerodynamic power $\langle P^* \rangle$ (solid) and force coefficient C_{F^*} (dashed). Dotted lines mark the $\pm 90°$ control bounds.

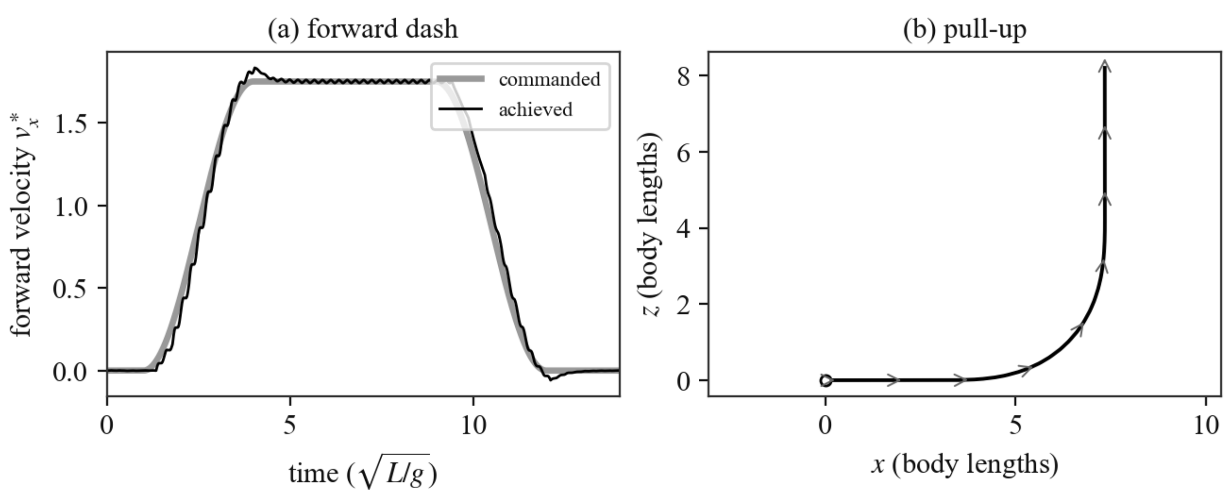

Figure 13: Closed-loop maneuvers under the generalized controller (unsteady simulation, control updated once per wingbeat). (a) Forward dash: commanded versus achieved forward velocity. (b) Pull-up: the sagittal plane path, with body velocity arrows rotating from horizontal to vertical.

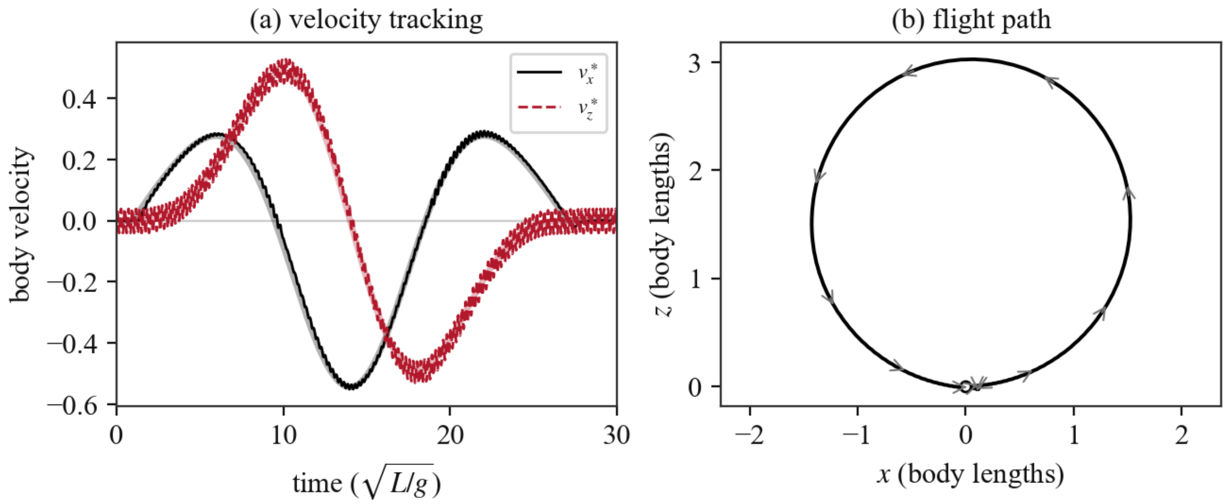

Figure 14: Closed-loop circular maneuver under the generalized controller (unsteady simulation). (a) Body velocity components v_x^* and v_z^*, commanded (light) versus achieved. Both reverse sign twice. (b) The sagittal plane flight path with body velocity arrows.

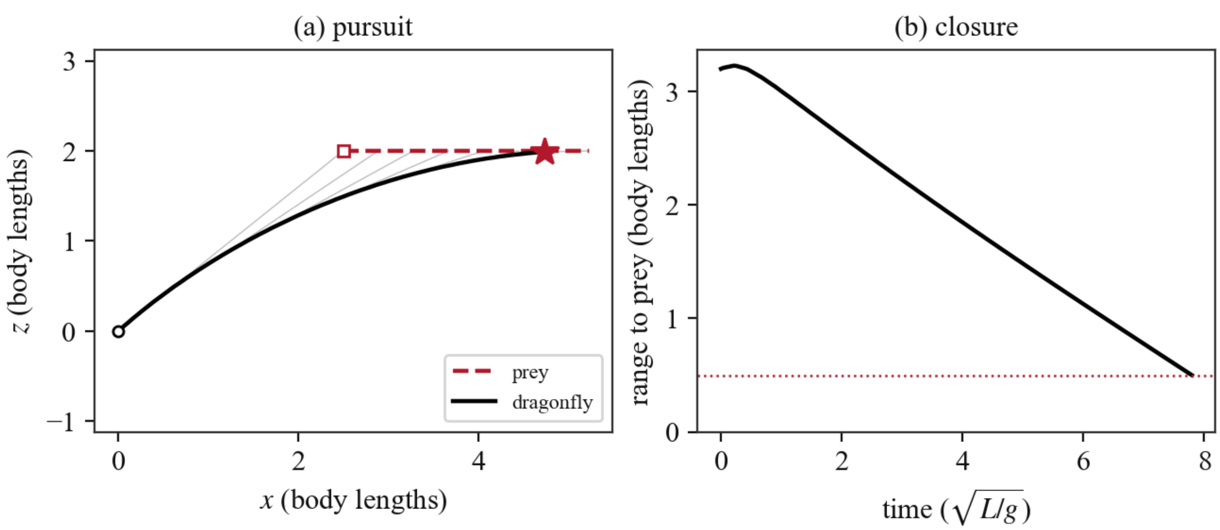

Figure 15: Pure-pursuit prey capture under the generalized controller. (a) Sagittal plane paths of the dragonfly (from rest at the origin) and the prey (moving horizontally), with line-of-sight rays. The star marks interception at range $0.5\,L$. (b) Range to the prey versus time.

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